$\boxed{\text{inverse hyperbolic functions}}$

0.

0.1 $\cosh^2(x) - \sinh^2(x) = 1$

proof: $\because \cosh(x) = \frac{e^x + e^{-x}}{2}$, $sh(x) = \frac{e^x - e^{-x}}{2}$

$\therefore ch^2 = \frac{e^{2x} + e^{-2x} + 2}{4}$, $sh^2 = \frac{e^{2x} + e^{-2x} - 2}{4}$

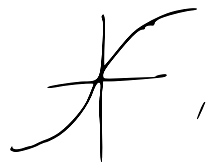
$\therefore ch^2 - sh^2 = 1$

0.2 $\because ch^2 - sh^2 = 1$

$\therefore 1 - \tanh^2(x) = \frac{1}{ch^2} = \operatorname{sech}^2(x)$

$\coth^2(x) - 1 = \frac{1}{sh^2} = \operatorname{csch}^2(x)$

$y = \sinh^{-1}(x)$, $x, y \in R$

1.2  , odd, $x, y \in R$

1.3 $y = \sinh^{-1}(x) = \ln(x + \sqrt{x^2+1})$, $\forall x \in R$

proof: $\because x = shy = \frac{e^y - e^{-y}}{2} = \frac{u - \frac{1}{u}}{2}$, $u = e^y$

$\therefore u^2 - 2xu - 1 = 0$, $(u-x)^2 = x^2 + 1$

$\therefore e^y = u = x \pm \sqrt{x^2+1}$

$\because x - \sqrt{x^2+1} < 0$, $\forall x \in R$

$\Leftrightarrow x^2 < x^2 + 1$, $\forall x > 0$

又 $\because e^y > 0$

$\therefore e^y = x + \sqrt{x^2+1}$

$\therefore y = \ln(x + \sqrt{x^2+1})$, $\forall x \in R$

1.4 $y = \sinh^{-1}(x) \Rightarrow y' = \frac{1}{\sqrt{x^2+1}}$, $\forall x \in R$

proof: $\because x = shy$

$\therefore 1 = chy \frac{dy}{dx}$, $\therefore \frac{dy}{dx} = \frac{1}{chy}$

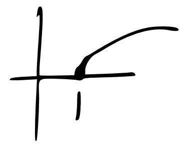
$\because chy = \pm\sqrt{sh^2y + 1} = \pm\sqrt{x^2+1}$

$\therefore \frac{dy}{dx} = \pm\frac{1}{\sqrt{x^2+1}}$

$\because \frac{dy}{dx} > 0$ $\therefore \frac{dy}{dx} = \frac{1}{\sqrt{x^2+1}}$, $\forall x \in R$

2. $y = \cosh^{-1}(x)$, $x \in [1, \infty)$

$y \in [0, \infty)$

2.2 

2.3 $y = \cosh^{-1}(x) = \ln(x + \sqrt{x^2-1})$

$\forall x \geq 1$

proof: $\because x = \cosh(y) = \frac{e^y + e^{-y}}{2}$

$\therefore x = \frac{u + \frac{1}{u}}{2}$, $u = e^y$

$\therefore u^2 - 2xu + 1 = 0$

$\therefore (u-x)^2 = x^2 - 1$

$\therefore e^y = u = x \pm \sqrt{x^2-1}$

$\therefore y = \ln(x \pm \sqrt{x^2-1})$

$\because x - \sqrt{x^2-1} < 1$, if $x > 1$

$\Leftrightarrow x - 1 < \sqrt{x^2-1} \Leftrightarrow (x-1)^2 < x^2 - 1$

$\Leftrightarrow x^2 + 1 - 2x < x^2 - 1 \Leftrightarrow x > 1$

$\therefore \ln(x - \sqrt{x^2-1}) < 0$, if $x > 1$

$\because y > 0$ if $x > 1$, $\therefore y = \ln(x + \sqrt{x^2-1})$

2.4 $y = \cosh^{-1}(x) \Rightarrow y' = \frac{1}{\sqrt{x^2-1}}$, $\forall x > 1$

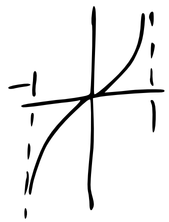
proof: $\because x = chy$ $\therefore 1 = shy \frac{dy}{dx}$, $\therefore \frac{dy}{dx} = \frac{1}{shy}$

$\because ch^2y - sh^2y = 1$, $\therefore shy = \pm\sqrt{ch^2y - 1} = \pm\sqrt{x^2-1}$

$\therefore \frac{dy}{dx} = \pm\frac{1}{\sqrt{x^2-1}}$ $\because \frac{dy}{dx} > 0$, $\forall x > 1$

$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{x^2-1}}$, $\forall x > 1$

3. $y=\tanh^{-1}(x)$

3.2  , odd, $x\in(-1,1)$

$y\in\mathbb{R}$

3.3 $y=\tanh^{-1}(x)=\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$

$\forall x\in(-1,1)$

proof: $\because x=\tanh y=\frac{\operatorname{sh}y}{\operatorname{ch}y}=\frac{e^y-e^{-y}}{e^y+e^{-y}}$

$=\frac{u-\frac{1}{u}}{u+\frac{1}{u}}$, $u=e^y$

$\therefore x=\frac{u^2-1}{u^2+1}$, $\therefore u^2(x-1)+x+1=0$

$\therefore u^2=\frac{1+x}{1-x}$ $\therefore e^y=u=\sqrt{\frac{1+x}{1-x}}$

$\therefore y=\frac{1}{2}\ln\frac{1+x}{1-x}$

3.4 $y=\tanh^{-1}(x)\Rightarrow y'=\frac{1}{1-x^2}$, $\forall x\in(-1,1)$

proof: $\because x=\tanh(y)$

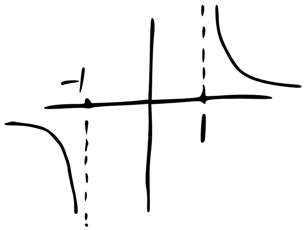
$\therefore 1=\tanh'(y)\frac{dy}{dx}$

$\therefore \frac{dy}{dx}=\frac{1}{\tanh'(y)}=\cosh^2(y)$

$\therefore \frac{dy}{dx}=\frac{1}{\operatorname{sech}^2(y)}=\frac{1}{1-\tanh^2(y)}$

$=\frac{1}{1-x^2}$

4. $y=\coth^{-1}(x)=\frac{1}{2}\ln\frac{x+1}{x-1}$

4.2  , odd, $x\in(-\infty,-1)\cup(1,\infty)$

$y\in\mathbb{R}\setminus\{0\}$

4.3 proof: $\because y=\coth^{-1}(x)$

$\therefore x=\coth(y)=\frac{1}{\tanh(y)}$

$\therefore \tanh(y)=\frac{1}{x}$

$\because \frac{1}{x}\in(-1,1)$, $\therefore y=\tanh^{-1}\left(\frac{1}{x}\right)$

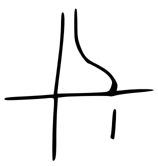
$\therefore y=\frac{1}{2}\ln\left(\frac{1+\frac{1}{x}}{1-\frac{1}{x}}\right)=\frac{1}{2}\ln\left(\frac{x+1}{x-1}\right)$

4.4 $y=\coth^{-1}(x)\Rightarrow y'=\frac{1}{1-x^2}$, $\forall |x|>1$

proof: $\because \frac{dy}{dx}=\frac{1}{\coth'(y)}=-\frac{1}{\operatorname{csch}^2(y)}$

$\therefore \frac{dy}{dx}=\frac{1}{1-\coth^2(y)}=\frac{1}{1-x^2}$

5. $y=\operatorname{sech}^{-1}(x)=\ln\left(\frac{1+\sqrt{1-x^2}}{x}\right)$, $\forall x\in(0,1]$

5.2  , $x\in(0,1]$, $y\in[0,\infty)$

5.3 proof: $\because y=\operatorname{sech}^{-1}(x)$

$\therefore x=\operatorname{sech}(y)=\frac{1}{\cosh y}$

$\therefore \cosh y=\frac{1}{x}$

$\because \frac{1}{x}\in[1,\infty)$

$\therefore y=\cosh^{-1}\left(\frac{1}{x}\right)=\ln\left(\frac{1}{x}+\sqrt{\frac{1}{x^2}-1}\right)$

$=\ln\left(\frac{1+\sqrt{1-x^2}}{x}\right)$, $\forall x\in(0,1]$

5.4 $y'=-\frac{1}{x\sqrt{1-x^2}}$, $\forall x\in(0,1)$

proof: $\because x=\operatorname{sech}(y)$, $\therefore \frac{dy}{dx}=\frac{1}{\operatorname{sech}'(y)}=-\frac{1}{\operatorname{sech}(y)\tanh(y)}$

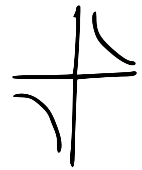
$\because \tanh(y)=\pm\sqrt{1-\operatorname{sech}^2(y)}$

$\therefore \frac{dy}{dx}=\pm\frac{1}{x\sqrt{1-x^2}}$, $\forall x\in(0,1)$

$\because x=\operatorname{sech}(y)\downarrow$, $y>0$. $\therefore y=\operatorname{sech}^{-1}(x)\downarrow$

$\therefore y'<0$ $\therefore \frac{dy}{dx}=-\frac{1}{x\sqrt{1-x^2}}$, $\forall x\in(0,1)$

6 $y = \operatorname{csch}^{-1}(x) = \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2}+1}\right), \forall x \neq 0$

6.2  , odd, $x, y \in \mathbb{R} \setminus \{0\}$

6.3 proof: $\because y = \operatorname{csch}^{-1}(x)$

$\therefore x = \operatorname{csch}(y) = \frac{1}{\sinh(y)}$

$\therefore \sinh y = \frac{1}{x}$,

$\therefore y = \sinh^{-1}\left(\frac{1}{x}\right)$

$= \ln\left(\frac{1}{x} + \sqrt{\frac{1}{x^2}+1}\right), \forall x \neq 0$

$= \ln\left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|}\right)$

6.4 $y' = -\frac{1}{|x|\sqrt{1+x^2}}, \forall x \neq 0$

proof: $\because x = \operatorname{csch}(y)$

$\therefore \frac{dy}{dx} = \frac{1}{\operatorname{csch}'(y)} = -\frac{1}{\operatorname{csch}(y)\coth(y)}$

$\because \coth(y) = \pm\sqrt{\operatorname{csch}^2(y)+1}$

$\therefore \frac{dy}{dx} = \pm\frac{1}{x\sqrt{x^2+1}}, \forall x \neq 0$

$\because \frac{dy}{dx} < 0$

$\therefore \frac{dy}{dx} = \begin{cases} -\frac{1}{x\sqrt{\ }}, & x>0 \\ \frac{1}{x\sqrt{\ }}, & x<0 \end{cases}$

$= -\frac{1}{|x|\sqrt{x^2+1}}, \forall x \neq 0$